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Melvin Immanuel Gregory

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AI Engineer | ML Engineer | Data Scientist

PROFESSIONAL SUMMARY

AI and ML engineer with a mathematics foundation (B.Sc. Top 5%) and an M.Sc. in Data Science in progress at VIT. Built and benchmarked agentic LLM systems in LangGraph, cutting LLM tokens per answer by 81% and raising routing accuracy from 72% to 94%, plus GAN forecasting models tested against classical baselines. Evaluating AI model reasoning at Handshake AI.

TECHNICAL SKILLS

LLMs & Agents	LangGraph, LangChain, RAG, Agentic Workflows, Prompt Engineering, LLM Evaluation, Ollama
Retrieval	Hybrid Search (BM25 + Dense), Reciprocal Rank Fusion, Embeddings, Semantic Search, Vector Search
Document AI	PDF/DOCX Parsing, Clause Extraction, EasyOCR, OCR, Document Extraction
Programming	Python, SQL, R
ML & Deep Learning	PyTorch, TensorFlow, Keras, Scikit-learn, GANs, BiLSTM, Hugging Face Transformers
Tools & Serving	FastAPI, Flask, Starlette, Git/GitHub, Jupyter, Google Colab
Data	Pandas, NumPy, SciPy, Matplotlib, Seaborn, Power BI

EXPERIENCE

Handshake AI – AI Analyst *Jul 2026 – Present*

- Evaluate AI-generated outputs for factual accuracy, logical reasoning, relevance, and instruction adherence.
- Contribute to multiple STEM projects, assessing the mathematical reasoning and problem-solving of AI models.
- Perform structured error analysis and model evaluation to identify failure patterns and improve AI response reliability.

Schneider Electric – Data Analytics Intern *Dec 2024 – Jan 2025*

- Developed interactive Power BI dashboards to visualize business KPIs and support data-driven decisions.
- Automated Excel reporting workflows with VBA macros, reducing repetitive manual reporting tasks.
- Performed data cleaning, transformation, and analysis using Excel and Power Query.

PROJECTS

LexAgent: Legal Document Intelligence Using Agentic AI GitHub

- Built an agent that answers questions on employment contracts and HR policies under Indian employment law (11 Acts, 492 sections), citing both the document clause and the statute section.
- Architected and benchmarked two LangGraph pipelines on identical inputs: an Agentic RAG baseline (hybrid BM25 + dense retrieval with reciprocal rank fusion, LLM routing and fact-checking) and a typed decision model navigating an Open Knowledge Format knowledge base (JEV + OKF).
- Cut LLM tokens per answer by 81% (8,264 to 1,590) and LLM calls from 4.9 to 0.8, raised routing accuracy from 72% to 94% and expected-section retrieval from 83% to 92%.
- Validated on 13 held-out questions over real documents and reported trade-offs openly; built a local web app with clickable citations using open models (Qwen2.5-7B via Ollama).

Adversarial Time-Series Forecasting Using GANs GitHub

- Compared a shared GAN generator against task-specific GANs (BiLSTM generators, WGAN-GP critic) for imputation and 7-day forecasting on 76 years of ERA5 climate data.
- Task-specific GAN reached R^2 0.85; the shared WGAN-GP gave 12% lower imputation error with 28% fewer generator parameters.
- Benchmarked against five classical baselines: ARIMA and ridge regression matched the GAN (R^2 0.86), a reminder to justify model complexity with evidence.

EDUCATION

M.Sc. Data Science – Vellore Institute of Technology (VIT), Chennai *Jun 2025 – Jun 2027*
CGPA: 9.02/10 · Coursework: Machine Learning, Deep Learning, Statistical Modeling, Data Mining, Big Data Analytics

B.Sc. Mathematics – Loyola College, Chennai *Jun 2022 – Jun 2025*
CGPA: 9.22/10 (Top 5%) · Coursework: Linear Algebra, Probability Theory, Statistical Inference, Regression Analysis

CERTIFICATIONS

- Stanford Online – Supervised Machine Learning: Regression and Classification
- Stanford Online – Unsupervised Learning, Recommenders and Reinforcement Learning